ADC

卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

https://blog.csdn.net/oushuwen/article/details/120968224

https://blog.csdn.net/wpgddt/article/details/150860400

我们使用的是S32K344，我们配置adc0/1/2 三个组都使用，每个组配置2个左右的通道用作测试

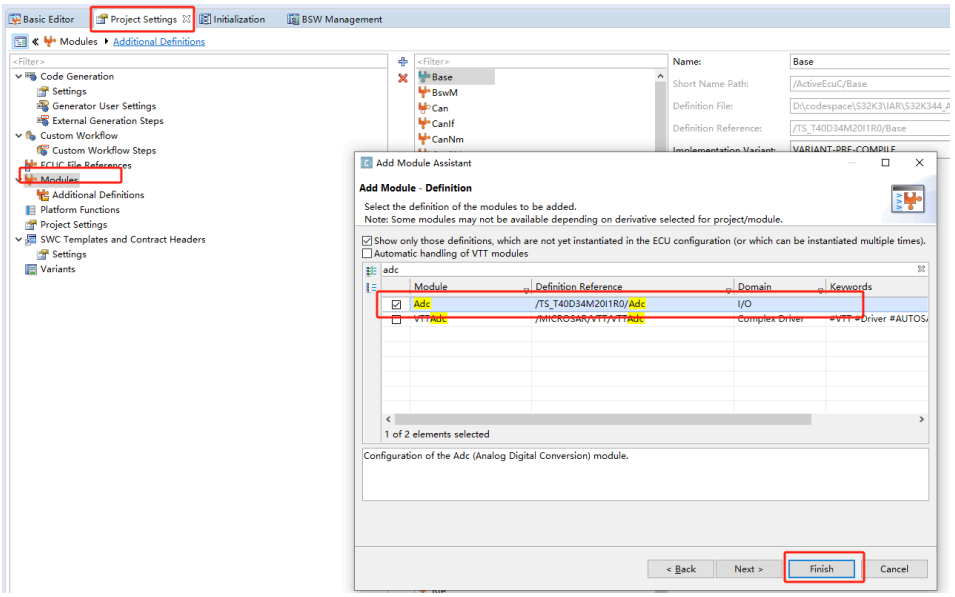
Adc0： PTD0(ADC0_P1)、PTD1(ADC0_P0)、PTD2(ADC0_S16)

Adc1：PTA12(ADC1_P0)、PTA11(ADC1_S10)

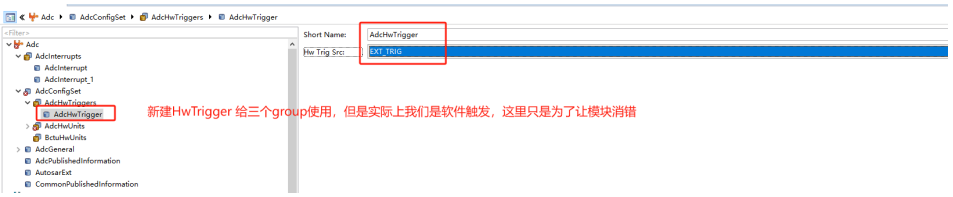
Adc2：PTA20(ADC2_P2)、PTA21(ADC2_S12)

一、davinci 添加Adc模块

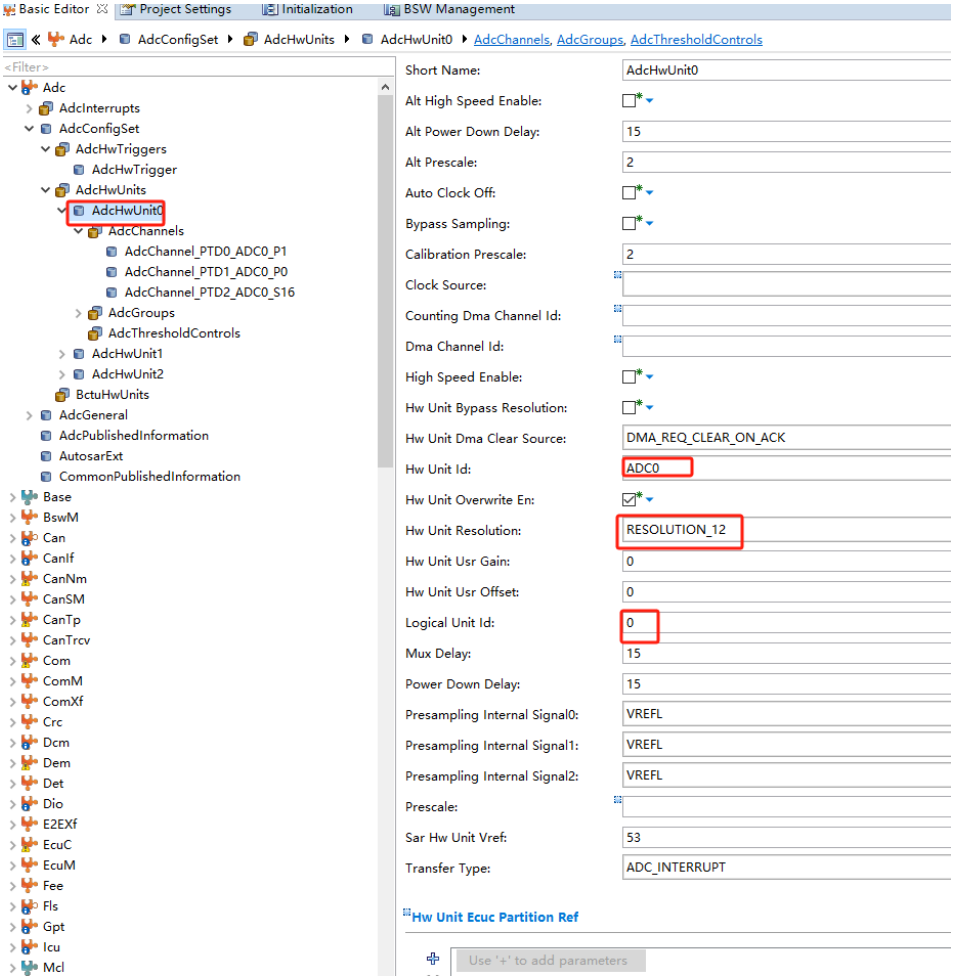
1.1、添加模块



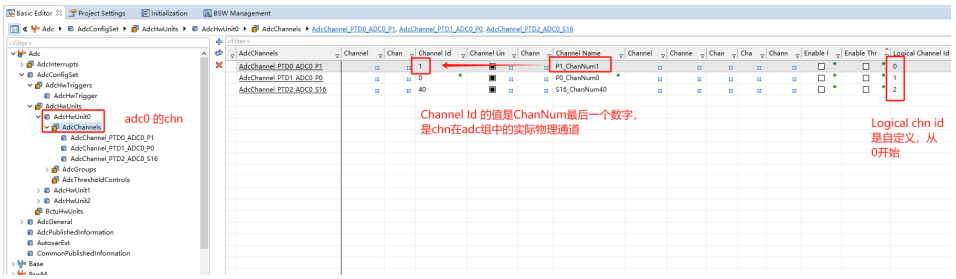
1.2、Adc模块消错



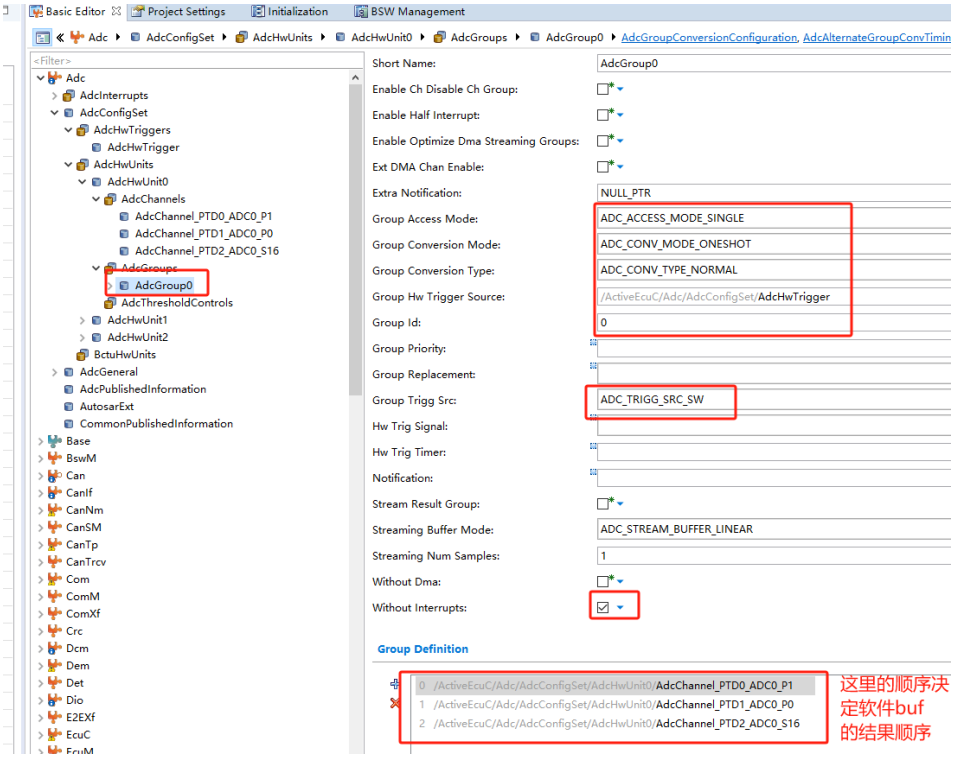
HwUnit0 , 我们配置12Bit



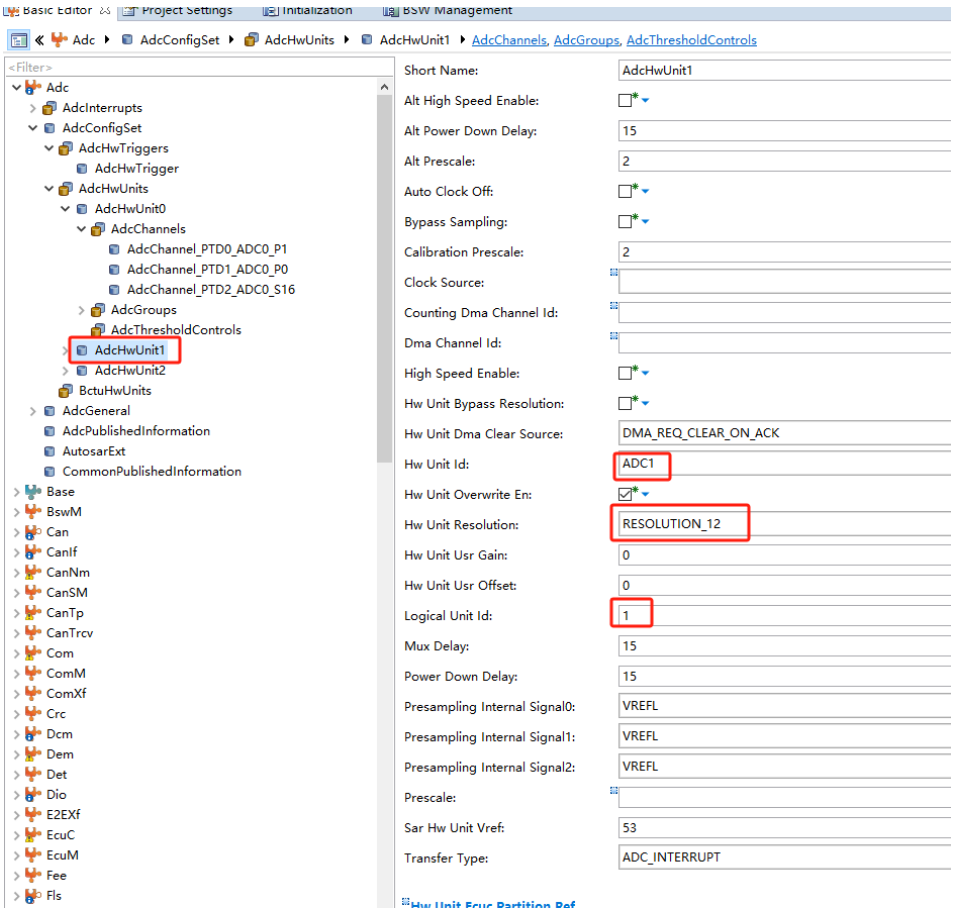
添加对应Adc0的对应chn

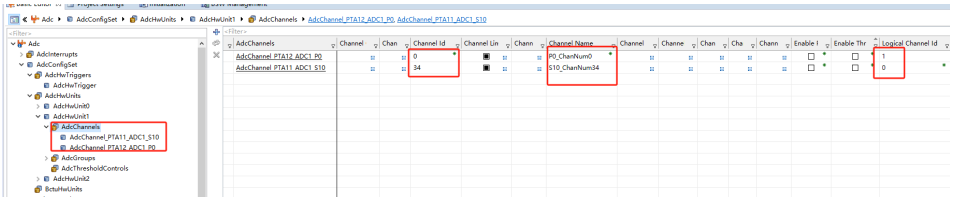


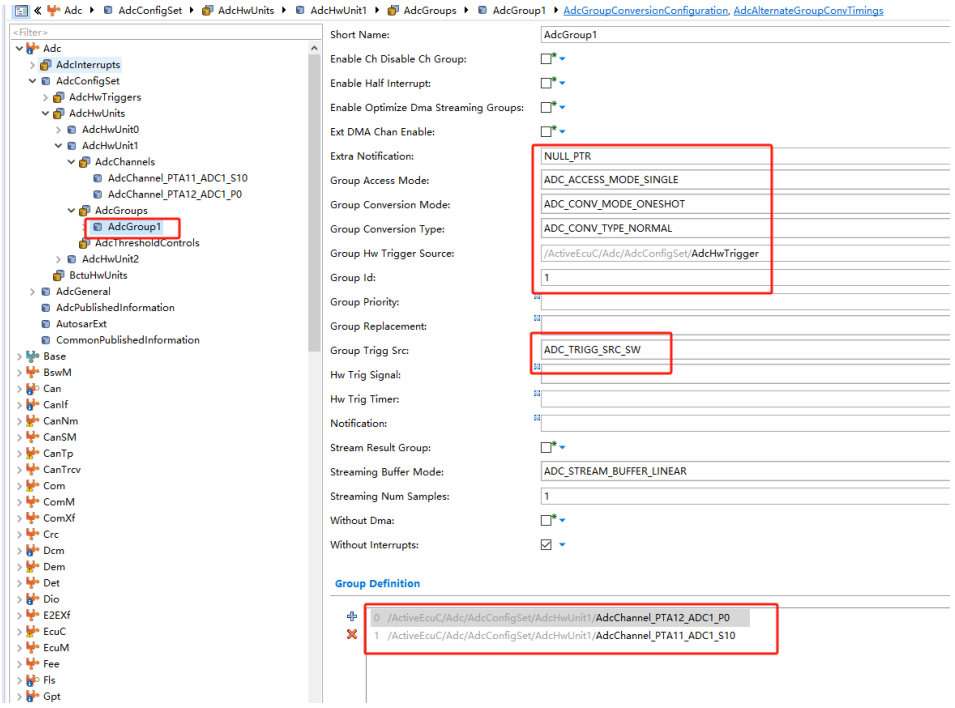
Group ， 配置单次采集，不使用中断方式


Basic Editor
Project Settings
Initialization
BSW Management
Adc ▸ AdcConfigSet ▸ AdcHwUnits ▸ AdcHwUnit0 ▸ AdcGroups ▸ AdcGroup0 ▸ AdcGroupConversionConfiguration, AdcAlternateGroupConvTimin
Short Name: AdcGroup0
Enable Ch Disable Ch Group:
Enable Half Interrupt:
Enable Optimize Dma Streaming Groups:
Ext DMA Chan Enable:
Extra Notification: NULL_PTR
Group Access Mode: ADC_ACCESS_MODE_SINGLE
Group Conversion Mode: ADC_CONV_MODE_ONESHOT
Group Conversion Type: ADC_CONV_TYPE_NORMAL
Group Hw Trigger Source: /ActiveEcuC/Adc/AdcConfigSet/AdcHwTrigger
Group Id: 0
Group Priority:
Group Replacement:
Group Trigg Src: ADC_TRIGG_SRC_SW
Hw Trig Signal:
Hw Trig Timer:
Notification:
Stream Result Group:
Streaming Buffer Mode: ADC_STREAM_BUFFER_LINEAR
Streaming Num Samples: 1
Without Dma:
Without Interrupts:
Group Definition
0 /ActiveEcuC/Adc/AdcConfigSet/AdcHwUnit0/AdcChannel_PTD0_ADC0_P1
1 /ActiveEcuC/Adc/AdcConfigSet/AdcHwUnit0/AdcChannel_PTD1_ADC0_P0
2 /ActiveEcuC/Adc/AdcConfigSet/AdcHwUnit0/AdcChannel_PTD2_ADC0_S16
这里的顺序决定软件buf的结果顺序

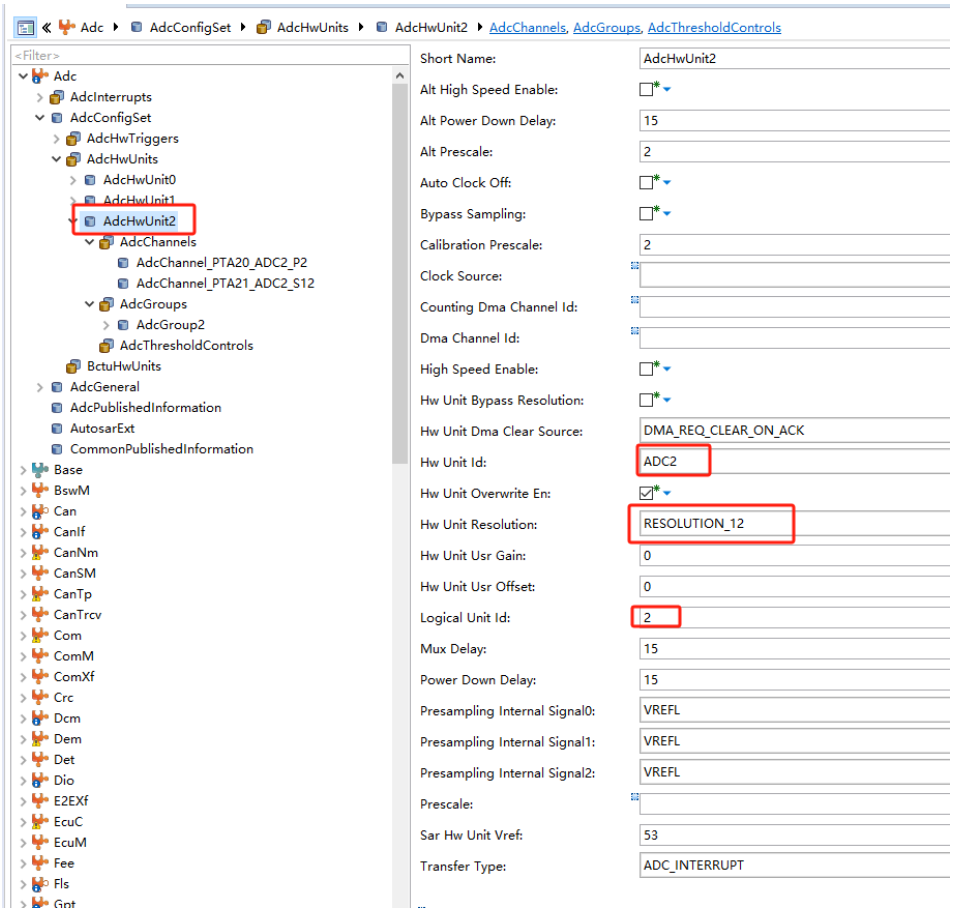
同理adc1

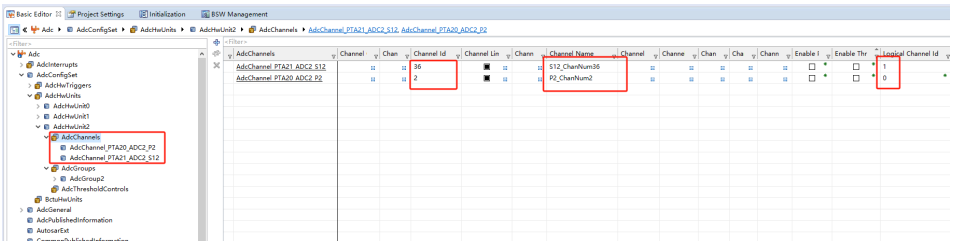

Basic Editor
Project Settings
Initialization
BSW Management
Adc ▸ AdcConfigSet ▸ AdcHwUnits ▸ AdcHwUnit1 ▸ AdcChannels, AdcGroups, AdcThresholdControls
Short Name: AdcHwUnit1
Alt High Speed Enable:
Alt Power Down Delay: 15
Alt Prescale: 2
Auto Clock Off:
Bypass Sampling:
Calibration Prescale: 2
Clock Source:
Counting Dma Channel Id:
Dma Channel Id:
High Speed Enable:
Hw Unit Bypass Resolution:
Hw Unit Dma Clear Source: DMA_REQ_CLEAR_ON_ACK
Hw Unit Id: ADC1
Hw Unit Overwrite En:
Hw Unit Resolution: RESOLUTION_12
Hw Unit Usr Gain: 0
Hw Unit Usr Offset: 0
Logical Unit Id: 1
Mux Delay: 15
Power Down Delay: 15
Presampling Internal Signal0: VREFL
Presampling Internal Signal1: VREFL
Presampling Internal Signal2: VREFL
Prescale:
Sar Hw Unit Vref: 53
Transfer Type: ADC_INTERRUPT

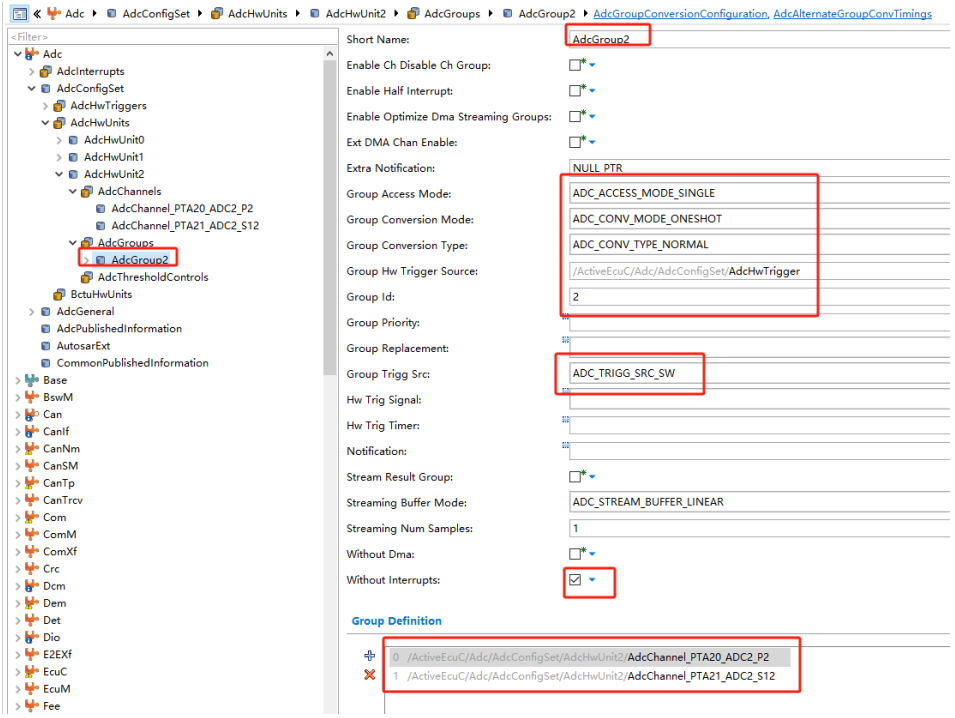

Adc ▸ AdcConfigSet ▸ AdcHwUnits ▸ AdcHwUnit1 ▸ AdcChannels ▸ AdcChannel_PTA12_ADC1_P0, AdcChannel_PTA11_ADC1_S10
AdcChannels
Channel Id
Channel Name
Logical Channel Id
AdcChannel_PTA12_ADC1_P0
AdcChannel_PTA11_ADC1_S10
0
34
P0_ChanNum0
S10_ChanNum34
1
0


Adc ▸ AdcConfigSet ▸ AdcHwUnits ▸ AdcHwUnit1 ▸ AdcGroups ▸ AdcGroup1 ▸ AdcGroupConversionConfiguration, AdcAlternateGroupConvTimings
Adc
AdcInterrupts
AdcConfigSet
AdcHwTriggers
AdcHwUnits
AdcHwUnit0
AdcHwUnit1
AdcChannels
AdcChannel_PTA11_ADC1_S10
AdcChannel_PTA12_ADC1_P0
AdcGroups
AdcGroup1
AdcThresholdControls
AdcHwUnit2
BctuHwUnits
AdcGeneral
AdcPublishedInformation
AutosarExt
CommonPublishedInformation
Base
BswM
Can
CanIf
CanNm
CanSM
CanTp
CanTrcv
Com
ComM
ComXf
Crc
Dcm
Dem
Det
Dio
E2EXf
EcuC
EcuM
Fee
Fls
Gpt
Short Name:
AdcGroup1
Enable Ch Disable Ch Group:
Enable Half Interrupt:
Enable Optimize Dma Streaming Groups:
Ext DMA Chan Enable:
Extra Notification:
NULL_PTR
Group Access Mode:
ADC_ACCESS_MODE_SINGLE
Group Conversion Mode:
ADC_CONV_MODE_ONESHOT
Group Conversion Type:
ADC_CONV_TYPE_NORMAL
Group Hw Trigger Source:
/ActiveEcuC/Adc/AdcConfigSet/AdcHwTrigger
Group Id:
1
Group Priority:
Group Replacement:
Group Trigg Src:
ADC_TRIGG_SRC_SW
Hw Trig Signal:
Hw Trig Timer:
Notification:
Stream Result Group:
Streaming Buffer Mode:
ADC_STREAM_BUFFER_LINEAR
Streaming Num Samples:
1
Without Dma:
Without Interrupts:
Group Definition
0 /ActiveEcuC/Adc/AdcConfigSet/AdcHwUnit1/AdcChannel_PTA12_ADC1_P0
1 /ActiveEcuC/Adc/AdcConfigSet/AdcHwUnit1/AdcChannel_PTA11_ADC1_S10

同理adc2


Adc ▸ AdcConfigSet ▸ AdcHwUnits ▸ AdcHwUnit2 ▸ AdcChannels, AdcGroups, AdcThresholdControls
<Filter>
Adc
AdcInterrupts
AdcConfigSet
AdcHwTriggers
AdcHwUnits
AdcHwUnit0
AdcHwUnit1
AdcHwUnit2
AdcChannels
AdcChannel_PTA20_ADC2_P2
AdcChannel_PTA21_ADC2_S12
AdcGroups
AdcGroup2
AdcThresholdControls
BctuHwUnits
AdcGeneral
AdcPublishedInformation
AutosarExt
CommonPublishedInformation
Base
BswM
Can
CanIf
CanNm
CanSM
CanTp
CanTrcv
Com
ComM
ComXf
Crc
Dcm
Dem
Det
Dio
E2EXf
EcuC
EcuM
Fee
Fls
Gpt
Short Name: AdcHwUnit2
Alt High Speed Enable:
Alt Power Down Delay: 15
Alt Prescale: 2
Auto Clock Off:
Bypass Sampling:
Calibration Prescale: 2
Clock Source:
Counting Dma Channel Id:
Dma Channel Id:
High Speed Enable:
Hw Unit Bypass Resolution:
Hw Unit Dma Clear Source: DMA_REQ_CLEAR_ON_ACK
Hw Unit Id: ADC2
Hw Unit Overwrite En:
Hw Unit Resolution: RESOLUTION_12
Hw Unit Usr Gain: 0
Hw Unit Usr Offset: 0
Logical Unit Id: 2
Mux Delay: 15
Power Down Delay: 15
Presampling Internal Signal0: VREFL
Presampling Internal Signal1: VREFL
Presampling Internal Signal2: VREFL
Prescale:
Sar Hw Unit Vref: 53
Transfer Type: ADC_INTERRUPT


Basic Editor
Project Settings
Initialization
BSW Management
AdcChannels
Channel Id
Channel Name
Logical Channel Id
AdcChannel_PTA21_ADC2_S12
AdcChannel_PTA20_ADC2_P2
36
2
S12_ChanNum36
P2_ChanNum2
1
0

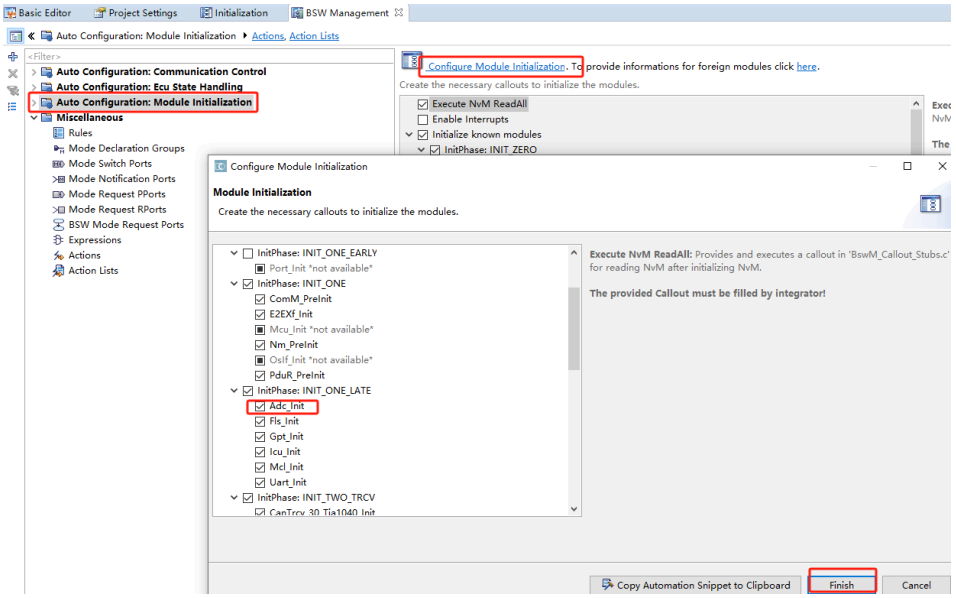


1.3、port模块说明

上述的port 都是由默认route的选择Adc， 所以port配置高阻态或者其他输入模式也可以使用Adc功能，如adc2 配置的两个， - 代表默认功能

PTA20	SIUL_MSCR20	0000_0000	GPIO[20]	SIUL		I/O	GPIO-STAND
		0000_0010	eMIOS_1_CH[2]_H	eMIOS_1	eMIOS Channel	O	
		0000_0100	LPSPI1_SIN	LPSPI1	LPSPI Serial Data Input	O	
		0000_0110	eMIOS_2_CH[2]_H	eMIOS_2	eMIOS Channel	O	
	-	-	ADC2_P2	ADC2	ADC Precise Input	I	
	-	-	WKPU[55]	WKPU	Wakeup Input	I	
	SIUL_IMCR530	0000_0010	EIRQ[2]	SIUL	External Interrupt	I	
	SIUL_IMCR594	0000_0010	eMIOS_1_CH[2]_H	eMIOS_1	eMIOS Channel	I	
	SIUL_IMCR626	0000_0010	eMIOS_2_CH[2]_H	eMIOS_2	eMIOS Channel	I	
	SIUL_IMCR751	0000_0011	LPSPI1_SIN	LPSPI1	LPSPI Serial Data Input	I	
	SIUL_IMCR870	0000_0001	TRGMUX_IN14	TRGMUX	Trigger Mux Input	I	
PTA21	SIUL_MSCR21	0000_0000	GPIO[21]	SIUL		I/O	GPIO-STAND
		0000_0001	LPSPI2_PCS2	LPSPI2	Peripheral Chip Select 2	O	
		0000_0010	eMIOS_1_CH[3]_H	eMIOS_1	eMIOS Channel	O	
		0000_0011	FXIO_D0	FXIO	FlexIO Bi-directional Shift/timer I/O	O	
		0000_0100	LPSPI1_PCS0	LPSPI1	Peripheral Chip Select 0	O	
		0000_0110	eMIOS_2_CH[3]_H	eMIOS_2	eMIOS Channel	O	
	-	-	ADC2_S12	ADC2	ADC Standard Input	I	
	SIUL_IMCR531	0000_0010	EIRQ[3]	SIUL	External Interrupt	I	
	SIUL_IMCR595	0000_0001	eMIOS_1_CH[3]_H	eMIOS_1	eMIOS Channel	I	
	SIUL_IMCR627	0000_0010	eMIOS_2_CH[3]_H	eMIOS_2	eMIOS Channel	I	
	SIUL_IMCR664	0000_0011	FXIO_D0	FXIO	FlexIO Bi-directional Shift/timer I/O	I	
	SIUL_IMCR744	0000_0011	LPSPI1_PCS0	LPSPI1	Peripheral Chip Select 0	I	
	SIUL_IMCR755	0000_0001	LPSPI2_PCS2	LPSPI2	Peripheral Chip Select 2	I	
	SIUL_IMCR871	0000_0001	TRGMUX_IN15	TRGMUX	Trigger Mux Input	I	

二、BswM添加adc_init

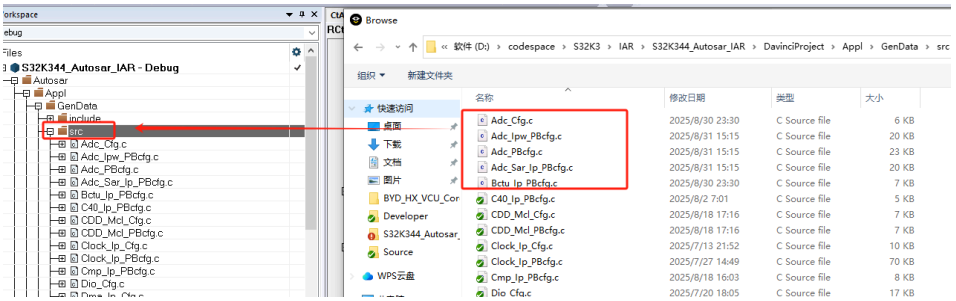


编译全部

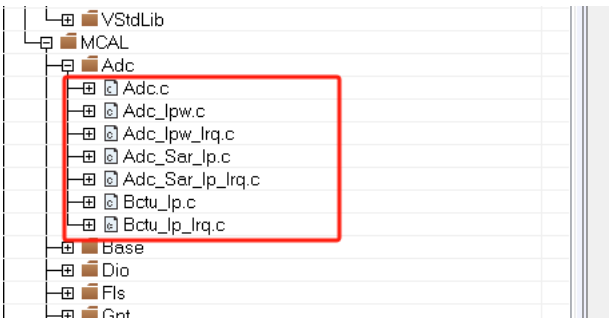
三、IAR 添加相关文件编译

3.1、添加编译文件

添加adc 生成的代码到GenData/src



添加MCAL的Adc，和头文件路径 到IAR 工程配置

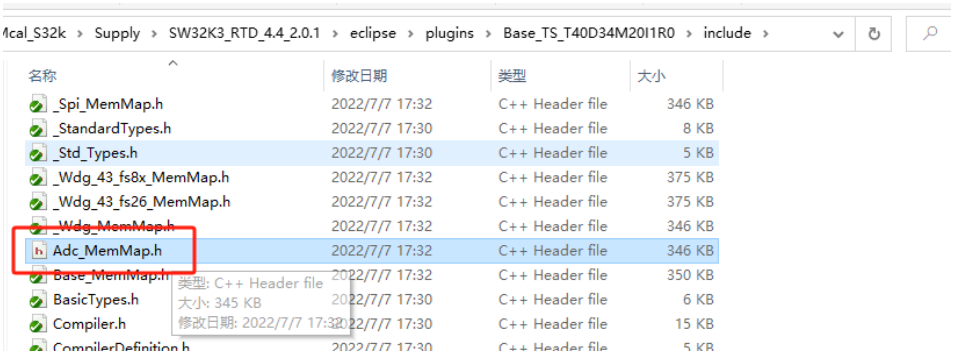


$PROJ_DIR$\CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Adc_TS_T40D34

3.2、使用Adc_MemMap.h

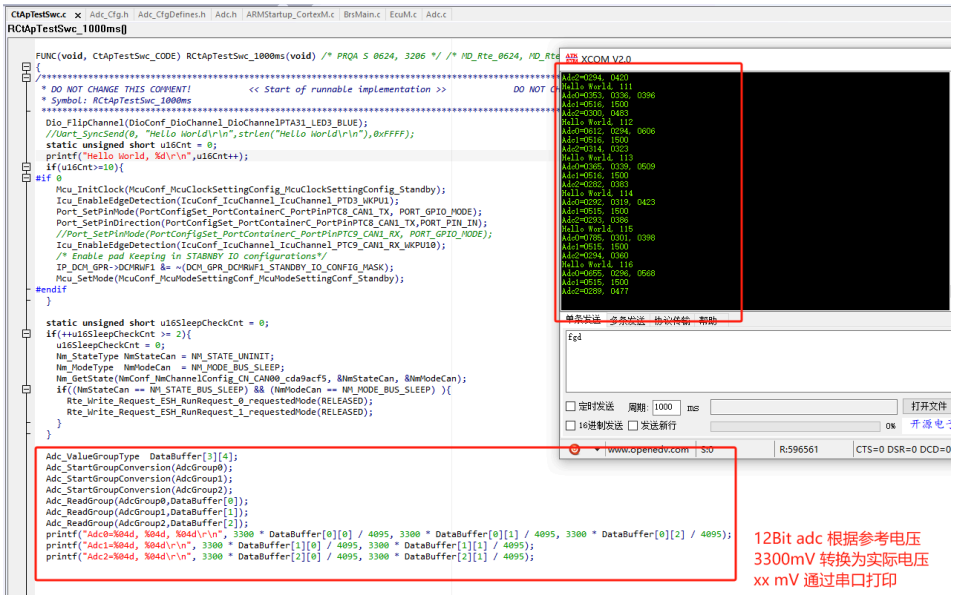
CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Base_TS_T40D34M20I1R0\inc
下

__Adc_MemMap.h 改为Adc_MemMap.h



四、添加测试代码

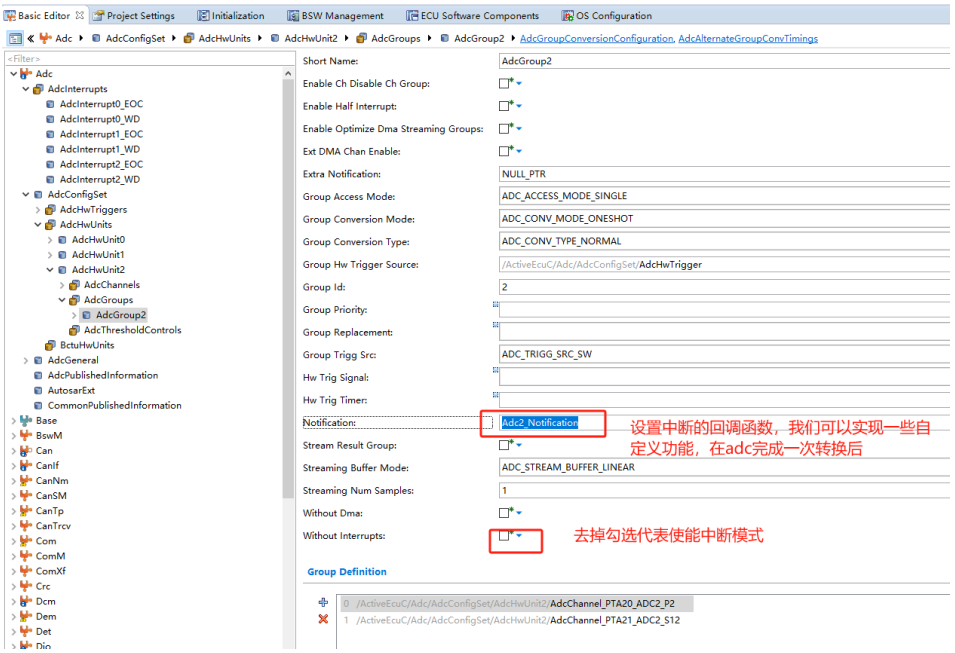
如下是配置不使用中断方式实现读取adc

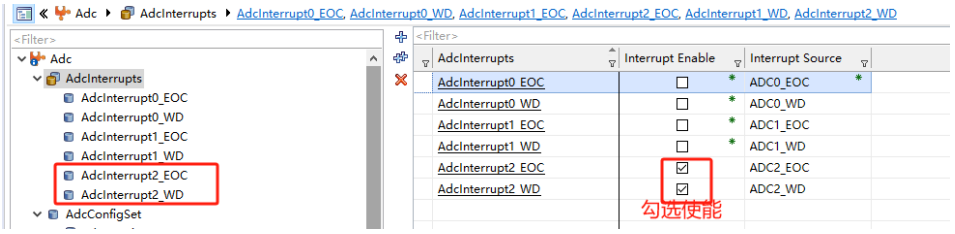


五、Adc 中断模式的配置及使用

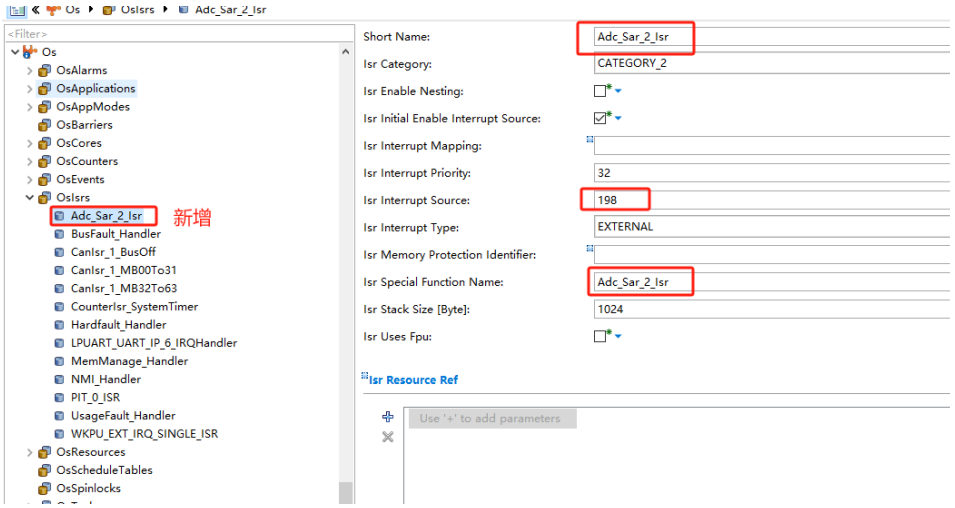
以配置Adc2 使用中断方式

5.1、Adc模块修改Adc2为中断模式





5.2、Os 模块添加Adc2的中断



中断函数名在mcal代码中找到

```
estSwc.c | Adc_Cfg.h | Adc_CfgDefines.h | Adc.h | ARMStartup_CortexM.c | BrsMain.c | EcuM.c | Adc.c | CtCdComplexDriver.c | CtApCanApp.c | Rte_CtApCanApp.h | Com_Cfg.h | CtApNvMSwc.c | Adc_Sar_Ip_Irq.c ×
r_Adc_Sar_2_Isr()
/* Check if Adc_Sar_Ip_Irq.c file and Adc_Sar_Ip_Irq.h file are of the same Autosar version */
#if ((ADC_SAR_IP_AR_RELEASE_MAJOR_VERSION_IRQ_C != ADC_SAR_IP_AR_RELEASE_MAJOR_VERSION_IRQ) || \
     (ADC_SAR_IP_AR_RELEASE_MINOR_VERSION_IRQ_C != ADC_SAR_IP_AR_RELEASE_MINOR_VERSION_IRQ) || \
     (ADC_SAR_IP_AR_RELEASE_REVISION_VERSION_IRQ_C != ADC_SAR_IP_AR_RELEASE_REVISION_VERSION_IRQ) \
    )
    #error "AutoSar Version Numbers of Adc_Sar_Ip_Irq.c and Adc_Sar_Ip_Irq.h are different"
#endif

/* Check if Adc_Sar_Ip_Irq.c file and Adc_Sar_Ip_Irq.h file are of the same Software version */
#if ((ADC_SAR_IP_SW_MAJOR_VERSION_IRQ_C != ADC_SAR_IP_SW_MAJOR_VERSION_IRQ) || \
     (ADC_SAR_IP_SW_MINOR_VERSION_IRQ_C != ADC_SAR_IP_SW_MINOR_VERSION_IRQ) || \
     (ADC_SAR_IP_SW_PATCH_VERSION_IRQ_C != ADC_SAR_IP_SW_PATCH_VERSION_IRQ) \
    )
  #error "Software Version Numbers of Adc_Sar_Ip_Irq.c and Adc_Sar_Ip_Irq.h are different"
#endif

/*==================================================================================================
*                                        GLOBAL CONSTANTS
==================================================================================================*/

/*==================================================================================================
*                                        GLOBAL FUNCTIONS
==================================================================================================*/
#define ADC_START_SEC_CODE
#include "Adc_MemMap.h"

ISR(Adc_Sar_0_Isr)
{
    /* Call IRQ handler with the correct instance */
    Adc_Sar_Ip_IRQHandler(0UL);
}

#if (ADC_INSTANCE_COUNT >= 2)
ISR(Adc_Sar_1_Isr)
{
    /* Call IRQ handler with the correct instance */
    Adc_Sar_Ip_IRQHandler(1UL);
}
#endif /* (ADC_INSTANCE_COUNT >= 2) */

#if (ADC_INSTANCE_COUNT >= 3)
ISR(Adc_Sar_2_Isr)
{
    /* Call IRQ handler with the correct instance */
    Adc_Sar_Ip_IRQHandler(2UL);
}
#endif /* (ADC_INSTANCE_COUNT >= 3) */

#define ADC_STOP_SEC_CODE
#include "Adc_MemMap.h"
```

中断号在 数据手册的 附件 中断.excel中查得

H979 | fx Error interrupt

	A	B	C	D	E	F	G	H
1	Address	Vector (all cores)	IRQ	NVIC IPR register number	NVIC non-IPR register number	INTM support	Instance	Interrupt request
960	0x000002E8	186	170	42	5	Y	LPSPI_5	LPSPI Interrupt
961	0x000002EC	187	171	42	5		RESERVED	
962	0x000002F0	188	172	43	5		RESERVED	
963	0x000002F4	189	173	43	5	Y	QSPI	TX Buffer Fill interrupt
964								Transfer Complete / Transaction Finished
965								RX Buffer Drain interrupt
966								Buffer Overflow / Underrun interrupt
967								Serial Flash Communication Error interrupt
968								All interrupts ORed output
969								FlashA ECC error
970	0x000002F8	190	174	43	5	Y	Synchronous Audio Interface 0	RX interrupt
971								TX interrupt
972	0x000002FC	191	175	43	5	Y	Synchronous Audio Interface 1	RX interrupt
973								TX interrupt
974	0x00000300	192	176	44	5		RESERVED	
975	0x00000304	193	177	44	5	Y	Ultra Secured Digital Host Controller	USDHC Interrupt
976	0x00000308	194	178	44	5	Y	JDC (JTAG)	Indicates new data to be read from JIN_IPS register or can be written to JOUT_
977	0x0000030C	195	179	44	5		RESERVED	
978	0x00000310	196	180	45	5	Y	ADC_0	End of conversion
979								Error interrupt
980								Watchdog interrupt
981	0x00000314	197	181	45	5	Y	ADC_1	End of conversion
982								Error interrupt
983								Watchdog interrupt
984	0x00000318	198	182	45	5	Y	ADC_2	End of conversion
985								Error interrupt
986								Watchdog interrupt
987	0x0000031C	199	183	45	5	Y	Low Power Comparator 0	Async interrupt
988	0x00000320	200	184	46	5	Y	Low Power Comparator 1	Async interrupt
989	0x00000324	201	185	46	5	Y	Low Power Comparator 2	Async interrupt
990	0x00000328	202	186	46	5		RESERVED	
991	0x0000032C	203	187	46	5		RESERVED	
992	0x00000330	204	188	47	5		RESERVED	

5.3、测试代码

```
 ***********************************************************************************************************/

FUNC(void, CtCdComplexDriver_CODE) RCtCdComplexDriver_Init(void) /* PRQA S 0624, 3206 */ /* MD_Rte_0624, MD_Rte_3206 */
{
/*********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>            DO NOT CHANGE THIS COMMENT!
 * Symbol: RCtCdComplexDriver_Init
 *********************************************************************************************************/
  printf("Swc Init, Reset Reason:%d\r\n",Mcu_GetResetReason());
  printf("Wakeup Src: WISR = 0x%08x, WISR_64= 0x%08x\r\n", u32WKPU_SR[0],u32WKPU_SR[1]);//IP_WKPU->WISR, IP_WKPU->WISR_64);

  Gpt_EnableNotification(GptConf_GptChannelConfiguration_GptChannelConfiguration_PIT0_CH0);
  Gpt_StartTimer(GptConf_GptChannelConfiguration_GptChannelConfiguration_PIT0_CH0, 40000000);

  Icu_EnableNotification(IcuConf_IcuChannel_IcuChannel_PTA2_WKPU0);
  Icu_EnableWakeup(IcuConf_IcuChannel_IcuChannel_PTA2_WKPU0);
  Icu_EnableEdgeDetection(IcuConf_IcuChannel_IcuChannel_PTA2_WKPU0);

  Adc_SetupResultBuffer(AdcGroup2, AdcDataBuffer[2]);
  Adc_EnableGroupNotification(AdcGroup2);

  //Icu_EnableWakeup(IcuConf_IcuChannel_IcuChannel_PTC9_CAN1_RX);
  /*Icu_EnableEdgeDetection(IcuConf_IcuChannel_IcuChannel_PTC9_CAN1_RX);*/

/*********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>              DO NOT CHANGE THIS COMMENT!
 *********************************************************************************************************/
}

#define CtCdComplexDriver_STOP_SEC_CODE
```

设置结果buffer，转换后结果直接存放，不需要调用readgroup

同时使能中断回调函数

CtApTestSwc.c | Adc_Cfg.h | Adc_CfgDefines.h | Adc.h | ARMStartup_CortexM.c | BrsMain.c | EcuM.c | Adc.c | CtCdComplexDriver.c | CtApCanApp.c | Rte_CtApCanApp.h | Com_Cfg.h | CtApNvM

RCtCdComplexDriver_500ms()

```
 * Executed if at least one of the following trigger conditions occurred:
 *   - triggered on TimingEvent every 500ms
 *
 **********************************************************************************************************/
/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!           << Start of documentation area >>                DO NOT CHANGE THIS COMMENT!
 * Symbol: RCtCdComplexDriver_500ms_doc
 **********************************************************************************************************/


/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!           << End of documentation area >>                  DO NOT CHANGE THIS COMMENT!
 **********************************************************************************************************/

FUNC(void, CtCdComplexDriver_CODE) RCtCdComplexDriver_500ms(void) /* PRQA S 0624, 3206 */ /* MD_Rte_0624, MD_Rte_3206 */
{
/**********************************************************************************************************
 * DO NOT CHANGE THIS COMMENT!           << Start of runnable implementation >>           DO NOT CHANGE THIS COMMENT!
 * Symbol: RCtCdComplexDriver_500ms
 **********************************************************************************************************/

 Adc_StartGroupConversion(AdcGroup0);
 Adc_StartGroupConversion(AdcGroup1);
 Adc_StartGroupConversion(AdcGroup2);
 Adc_ReadGroup(AdcGroup0,AdcDataBuffer[0]);
 Adc_ReadGroup(AdcGroup1,AdcDataBuffer[1]);
 //Adc_ReadGroup(AdcGroup2,AdcDataBuffer[2]);
 //printf("Adc0=%04d, %04d, %04d\r\n", 3300 * AdcDataBuffer[0][0] / 4095, 3300 * AdcDataBuffer[0][1] / 4095, 3300 * AdcDataBuffer[0][2] / 4095);
 //printf("Adc1=%04d, %04d\r\n", 3300 * AdcDataBuffer[1][0] / 4095, 3300 * AdcDataBuffer[1][1] / 4095);
 //printf("Adc2=%04d, %04d\r\n", 3300 * AdcDataBuffer[2][0] / 4095, 3300 * AdcDataBuffer[2][1] / 4095);
```

启动转换还是要的， ReadGroup不用了

```
void Adc2_Notification(void)
{
  if (ADC_STREAM_COMPLETED == Adc_GetGroupStatus(AdcGroup2))
  {
    //add your code when adc Conversion finished
    //Adc_ReadGroup(AdcGroup2,AdcDataBuffer[2]);
  }
}
```